

Open source libraries, frameworks and tools to implement machine learning

In order to implement machine learning on a system for any scenario, there are enough open source tools, software or frameworks available for us to choose from, based on our preference for a specific language or environment. Let's have a look at some of them.

Shogun: Shogun is one of the oldest and most venerable of the different machine learning libraries available in the market. It was first developed in 1999 using C++, but now it's not limited to working in C++ only; rather, it can be used transparently in many languages and environments such as Java, C#, Python, Ruby, R, Octave, Lua and MATLAB. It is very easy to use, and quite fast at compilation and execution. As it can be used in a wide range of languages, people feel comfortable with it.

Weka: Weka was developed at the University of Waikato in New Zealand. It collects a set of Java machine learning algorithms which are engineered specifically for data mining. This GNU GPLv3-licensed collection possesses a package system, which can be used to extend its functionality. It has both official and unofficial packages available. Weka comes with a book that explains both the software and the techniques used in it. While Weka is not aimed specifically for Hadoop users, it can be used with Hadoop as well, because of the set of wrappers that have been produced for the most recent versions of it. It doesn't support Spark, but Clojure users can also use Weka.

CUDA-Convnet: CUDA-Convnet is a machine learning library especially used for neural-network applications. It is written in C++ in order to exploit Nvidia's CUDA GPU processing technology. It can even be used by those who prefer Python rather than C++. The resulting neural nets obtained as output from this library can be saved as Python pickled objects and, hence, can be accessed from Python.

We should note that the original version of the project is no longer being developed, but has been reworked into a successor named CUDA-Convnet2. It supports multiple GPUs and even Kepler-generation GPUs.

H2O: H2O is an open source machine learning framework developed by Oxdia. H2O's algorithms are basically geared for business processes, like fraud or trend predictions. H2O can easily interact in a standalone fashion with different HDFS stores. It can be in MapReduce, on top of YARN, or directly in an Amazon EC2 instance as well. Hadoop MapReduce can use Java for interaction with H2O, but this framework also provides bindings for R, Python and Scala. It enables cross-interaction with all the libraries that are available on those platforms.

The applications of machine learning in the current tech era

Guess we are all curious and interested to know the applications of machine learning in the current scenario

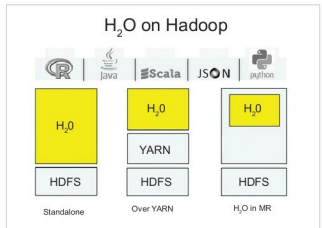


Figure 4: H2O on Hadoop (image credits: www.infoworld.com)

when the world is on the path to becoming smarter through the automation of all possible manual tasks. Google and Facebook use machine learning to push their respective advertisements to the relevant users. Here are a few applications that we should know of.

1. **Banking and financial services:** Machine learning is widely used to predict the customers who are likely to be defaulters in paying credit card bills or in repaying loans. This is of utmost importance as machine learning helps banks to identify the customers who can be given credit cards and loans.
2. **Healthcare:** It is widely used to diagnose various deadly diseases (like cancer) on the basis of patients' symptoms, and tallying them with the past data available for similar kinds of patients.
3. **Retail:** Machine learning is used to identify products that sell fast and those that don't. It helps retailers to decide on the kind of products to introduce or remove from their stock. Also, machine learning algorithms can be very effective in finding two or more products that will sell together. This is basically done to encourage customer loyalty initiatives which, in turn, help different retailers to develop and maintain loyal customers. Walmart, Amazon, Big Bazaar and all other such retail units extensively make use of machine learning.
4. **Publishing and social media:** There are different publishing firms like LexisNexis and Tata McGraw Hill, which make use of machine learning to run the queries and fetch the documents required by their users online, based on their preferences and requirements. Google and Facebook also use these techniques to rank their search outputs and news feeds. Facebook also provides a list of possible friends under its 'Friend suggestions' using this.
5. **Robot locomotion:** Robot locomotion is basically a collective term used for the different methods that robots use to transport themselves from one place to other. A major challenge in this field lies in developing capabilities for different robots to autonomously decide